INCH-POUND

MIL-E-62140A(AT)
18 December 1989
SUPERSEDING
MIL-E-62140(AT)
27 August 1971

MILITARY SPECIFICATION

ENGINE, DIESEL: LIQUID-COOLED, V-TYPE, 6-CYLINDER, 210 H.P.

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 210 horsepower, 6-cylinder, V-type, liquid-cooled, 2-stroke-cycle, internal-combustion, compression-ignition (diesel) engine for use in military vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2815

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SPECIFICATIONS

FEDERAL

O-A-548	- Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated.
P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty.
MIL-F-46162	- Fuel, Diesel, Referee Grade.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-A-53009	- Additive, Antifreeze Extender, Liquid Cooling System.

STANDARDS

MILITARY

MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packaging and Marking.
MIL-STD-1400	- Engines, Gasoline and Diesel, Methods of Test.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

8738127	- Engine (Complete).
11650202	- Containers, Shipping and Storage, 6V53 Engine.

PACKAGING DATA

57K1000	- Special Packaging Instruction - Engine and Container.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish engines which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent engines delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Material. Materials shall be as specified herein, on applicable standards or drawings, and in applicable specifications (see 4.8.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.8.1 and 6.6.2).

3.3 Design and construction. The engine shall be constructed in accordance with Drawing 8738127 (see 4.8.1 and 4.8.2).

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3.3.1 Accessories and equipment. Unless otherwise specified (see 6.2), all accessories and equipment shall be installed on the engine and properly adjusted. All electrical accessories and equipment, including wiring and electrical connections, shall conform to the applicable requirements of MIL-STD-1184 (see 4.8.1).

3.3.2 Interchangeability of parts. With the exception of parts furnished in material sets or for which a selective fit is specified, component assemblies, parts or service parts shall be so constructed as to facilitate ease of installation, replacement, and adjustment without requiring modification. Any design change that affects performance or changes the configuration of the above items requires Government approval (see 4.8.1).

3.3.3 Production break-in. The engine, using fuel oil conforming to grade 2 of VV-F-800, shall be given a break-in run by the contractor to ensure the engine, as submitted to the Government for acceptance, shall meet all requirements specified herein (see 4.8.3).

3.4 Performance. The engine performance requirements shall be met under the following conditions:

a. Lubricating oil. Lubricating oil conforming to the seasonal requirements of MIL-L-2104 from -10 to 115 degrees Fahrenheit (°F) and MIL-L-46167 from -65 to 0°F shall be used. The contractor may use oil of his own selection, or oil conforming to grade 2 of MIL-L-21260, during engine build-up and the break-in run (see 6.6.4).

b. Coolant. Water and antifreeze compound conforming to MIL-A-46153 heavy duty antifreeze or O-A-548, arctic-type antifreeze compound conforming to MIL-A-11755, or antifreeze extender additive conforming to MIL-A-53009 shall be used.

c. Rated operating conditions. The engine rating shall be corrected to the following conditions in accordance with MIL-STD-1400:

Dry air barometric pressure	29.00 inches mercury (in Hg)
Inlet air temperature	85°F
Vapor pressure	0.38 inch Hg
Dry air density	0.0705 pound per cubic foot.

3.4.1 Endurance. The engine shall retain 90 percent (%) of its rated horsepower after being subjected to a 400-hour endurance test without requiring any more maintenance than changing oil, replacing oil and fuel filters, and minor adjustments (see 4.8.4).

3.4.1.1 Endurance run fuel. During the endurance run, the engine shall be operated with referee-grade diesel fuel conforming to MIL-F-46162.

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3.4.1.2 Endurance run lubricating oil. During the endurance run, the engine shall be operated with reference grade lubricating oil conforming to MIL-L-2104 (see 6.6.4).

3.4.2 Speed range. The engine shall operate satisfactorily under all loads and conditions as specified herein through a speed range of 1200 to 2785 revolutions per minute (rpm). The engine shall maintain an idle speed of 650 to 700 rpm when using diesel fuel conforming to grade 2 of VV-F-800 (see 4.8.4).

3.4.2.1 Governor. The governor shall limit the engine speed as follows:

Full-load operation	2785 to 2835 rpm
No-load operation	2940 to 2990 rpm.

The engine speed with no load on the engine shall stabilize within 3 seconds after full-rack position is reached (see 4.8.2 and 4.8.4).

3.4.3 Brake horsepower (bhp). At the completion of the production break-in run, the engine operating at 2785 rpm and corrected to MIL-STD-1400 operating conditions, shall develop not less than 202 gross bhp (see 4.8.4 and 6.6.3).

3.4.4 Torque. The engine operating at 1600 rpm shall develop not less than 420 pound-feet observed torque using specified fuel at the completion of the production break-in run (see 4.8.4).

3.4.5 Exhaust smoke density. The exhaust smoke density at the full power position, when measured within 3 feet of the exhaust manifold outlet, shall not exceed the conditions specified in table I when using fuel specified herein (see 4.8.4 and 6.6.5).

TABLE I. Exhaust smoke density.

Engine speed (rpm)	Visual No.	Meter No. 1/
1000	3	8
1400	3	7
2600	1	4

1/ The meter reading shall have precedence over the visual reading.

3.4.6 Fuel consumption. The fuel consumption shall not exceed that specified in table II with the engine operating at 2785 rpm using fuel specified herein (see 4.8.4).

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TABLE II. Maximum observed brake specific fuel consumption.

Engine speed (rpm)	Pound per bhp-hour
1800	0.460
2785	0.450

3.4.7 Oil consumption. After 25 hours of operation, the engine shall consume not more than 0.007 pounds of lubricating oil per observed bhp-hour, when operating at full rack throughout the power check phase of the engine with an oil sump temperature of 180 to 260°F. The power check phase shall be determined at full rack position with the engine operating at full load at speeds of 1600, 2000, 2400, and 2785 rpm to assure meeting the performance requirements (see 4.8.4).

3.4.8 Oil pressure. Under all operating conditions specified herein, including a 1-quart-low oil sump condition, and with an oil sump temperature of not less than 180°F, the gallery oil pressure shall not be more than 80 pounds per square inch (psi) nor less than 40 psi when the engine is operating at 2785 rpm using specified oil, and shall not be less than 5 psi when the engine is idling at 650 to 700 rpm (see 4.8.4).

3.4.9 Grades and slopes. The engine shall start and demonstrate the performance characteristics specified herein on longitudinal grades and lateral slopes up to 60% in all directions, when operated throughout the speed range of the engine, with full and 1-quart-low oil level conditions in the sump. As a result of said operation, no evidence of faulty lubrication, cooling, fuel supply, leakage, or other malfunction shall be found (see 4.8.5).

3.4.10 Steam and water jet cleaning. The engine and all its components shall withstand cleaning with a high-pressure steam cleaner conforming to P-C-437 and water jet without deterioration of seals or hoses, water leakage past seals or gaskets, or other defects (paint removal shall not be considered a defect). The jet shall be applied perpendicular to the surface being cleaned at a distance of not less than 1 foot from the surface and cleaned at a rate of 1 square foot per minute. The jet pressure shall not be less than 100 psi, nor more than 110 psi (see 4.8.6).

3.4.11 Electromagnetic compatibility. Unless otherwise specified (see 6.2), the engine shall meet the functional suppression requirements in accordance with class C1 of MIL-STD-461 (see 4.8.7).

3.5 Environmental. The engine shall start within 2 minutes and shall demonstrate the performance characteristics specified herein under any of the following conditions or possible combination of conditions:

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3.5.1 Temperature (see 4.8.8.1).

a. Low. With integral cold starting aid, and without external aids after being cold-soaked, without benefit of solar radiation, to an ambient temperature of -25°F. Not less than 100 rpm cranking speed shall be demonstrated with 16 V dc available at the starter terminals. (Cold-soak shall be defined as reducing the temperature of the engine and its fuel, coolant, and lubricating oil to within 5°F of the ambient air temperature.)

b. High. Without external aids, and a fuel pressure of 5 psi at the inlet to the fuel injector supply pumps, when operated at the following temperature with exposure to maximum solar radiation:

Ambient air	115°F
Intake air (at air cleaner inlet)	115°F
Coolant outlet	230°F
Oil sump	At maximum temperature attainable up to 275°F.

3.5.2 Elevation. At any elevation from sea level to 8,000 feet, except that the power requirements shall be reduced proportional to the elevation, so that at the 8,000-foot elevation, the observed power output of the engine shall not be less than 75% of rated power (see 4.8.8.2).

3.5.3 Humidity. Under relative humidity conditions as low as 5% at a temperature of 115°F and as high as 100% at temperatures from -25° to 85°F (see 4.8.8.3).

3.6 Exterior surface treatment. All exposed exterior surfaces of the engine and its components shall be cleaned and painted or treated for corrosion resistance as specified on the applicable drawing, or if not so specified, in accordance with MIL-STD-193 (see 4.8.2).

3.7 Plates. Unless otherwise shown on the applicable drawings, the nameplate, data, and instruction plates shall conform to MIL-P-514 (see 4.8.2 and 6.2).

3.8 Workmanship. Workmanship shall be of the quality necessary to produce engines free from defects that would affect their serviceability (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
c. Control tests (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F.

b. Relative humidity: Uncontrolled room ambient.

c. Atmospheric pressure: Site pressure.

4.4 First article inspection. When first article is required (see 3.1), first article inspection shall be performed on first article samples (see 6.2 and 6.3). Unless otherwise specified (see 6.2), first article inspection shall include the inspections specified in table III.

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TABLE III. Classification of inspection.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Materials, design, and construction	3.2 thru 3.3.2	4.8.1	X			
Examination (see table IV)	3.3, 3.4.2.1, 3.6, 3.7 and 3.8	4.8.2	X	X		
Production break-in	3.3.3	4.8.3	X			
Endurance						
First article	3.4.1	4.8.4.1	X			
Control	3.4.1	4.8.4.2				X
Speed range	3.4.2	4.8.4	X		X	X
Governor	3.4.2.1	4.8.4	X		X	X
Bhp	3.4.3	4.8.4	X		X	X
Torque	3.4.4	4.8.4	X		X	X
Exhaust smoke density	3.4.5	4.8.4	X			X
Fuel consumption	3.4.6	4.8.4	X			X
Oil consumption	3.4.7	4.8.4	X			X
Oil pressure	3.4.8	4.8.4	X		X	X
Grades and slopes	3.4.9	4.8.5	X			
Steam and water jet cleaning	3.4.10	4.8.6	X			
Electromagnetic compatibility	3.4.11	4.8.7	X			
Temperature	3.5.1	4.8.8.1	X			
Elevation	3.5.2	4.8.8.2	X			
Humidity	3.5.3	4.8.8.3	X			

4.5 Quality conformance inspections (QCI). QCI shall be conducted on the sample engines and shall include the examination of 4.5.1 and the tests of 4.5.2 (see 6.2). Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.5.1 Examination. The sample engine shall be examined for the defects specified in table IV (see 6.2).

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TABLE IV. Classification of defects.

Category	Defect	Method of Examination
Major		
101	Exhaust valve tappet clearance - improper adjustment (see 3.3).	Gage
102	Oil sump; fuel and oil filters - dirt, chips, or foreign objects (see 3.3).	Visual and functional
103	Fuel, oil, and coolant - leakage (see 3.3).	Visual
104	Fuel lines - damaged (see 3.3).	Visual and functional
105	Fuel system components - malfunction, damaged, or leaks (see 3.3).	Visual and functional
106	Fuel injection system, timing, and components - malfunction, improper adjustment (see 3.3).	Gage and functional
107	Cylinder head bolts, intake and exhaust manifold flange bolts, and vibration damper bolts - improper torque (see 3.3).	Torque wrench
108	Crankshaft - excessive or restrictive end play (see 3.3).	Gage and functional
109	Crankshaft pressure - exceeds limits (see 3.3).	Gage
110	Air cleaner - malfunction (see 3.3).	Functional
111	Air box pressure - exceeds limits (see 3.3).	Gage
112	Governor - malfunction (see 3.4.2.1).	Visual and functional
113	Workmanship - faulty affecting performance (see 3.8).	Visual and functional
Minor		
201	Linkage - improper adjustment (see 3.3).	Visual
202	Minor assemblies - omitted (see 3.3).	Visual
203	Engine - improper adjustment or installation of components (see 3.3).	Visual and functional
204	Fuel lines, hose, vents, and shut-off valve - improper assembly or installation (see 3.3).	Visual
205	Exterior surface treatment - improper application (see 3.5).	Visual
206	Plates - improper (see 3.7).	Visual
207	Workmanship - faulty affecting appearance (see 3.8).	Visual

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4.5.2 QCI tests. Subsequent to the examination of 4.5.1, the sample engines shall be subjected to the QCI tests specified in table III.

4.6 Control tests. Sample engines selected for control tests shall be new and examined in accordance with 4.5.1 prior to subjecting the sample engines to the control tests specified in table III (see 4.8.4.2 and 6.2).

4.7 Inspection of packaging. The preservation, packaging, packing, and marking shall be inspected to determine conformance to 5.1.

4.8 Methods of inspection.

4.8.1 Materials, design and construction. Conformance to 3.2 through 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.4.2.1, 3.6, 3.7, and 3.8 shall be determined by examination for the defects listed in table IV.

4.8.3 Production break-in. To determine conformance to 3.3.3, the engine shall be operated as specified herein. Any unsatisfactory operation or conditions which require correction shall be corrected by the contractor before the engine is presented for acceptance.

4.8.3.1 Teardown inspection. At the conclusion of the break-in run, the contractor, in the presence of the Government, shall disassemble and inspect the first 10 engines, and every second engine of the next 10 of any contract except overlapping contracts. Disassembly shall be accomplished to the extent required to perform the following inspection:

a. Oil contamination.
b. Dirt, chips, or foreign matter in the engine block, oil pan filters, or accessory drive case.
c. Main bearing bolt torque. 1/
d. Connecting rod bolt torque. 1/
e. Cylinder head bolt torque. 1/
f. Cylinder bore scuffing, scoring, galling, etc.
g. Piston scuffing and burning.
h. Correct valve adjustment.
i. Correct rack and governor linkage adjustment.
j. Missing parts.

Any engine not requiring replacement parts shall be reassembled and subjected to the acceptance power check test only.

1/ Nuts and bolts listed by the manufacturer as requiring exceptions to standard bolt and nut torque specifications.

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4.8.3.1.1 Corrections and reassembly. If there is evidence of any condition of engine or parts requiring correction, such corrections shall be made by the contractor prior to reassembly of the engine and resubmission to the break-in run. In addition, the contractor shall submit objective evidence to the Government that corrective action has been taken in production to eliminate any deficiency disclosed. Failure of the contractor to submit such evidence shall be cause for refusal by the Government to continue acceptance of subsequent engines. Any engine requiring replacement parts shall be reassembled and subjected to the break-in run and acceptance power check test.

4.8.4 Endurance. To determine conformance to 3.4.1, the engine shall be subjected to the applicable test as follows.

4.8.4.1 First article.

4.8.4.1.1 NATO 400-hour test. Unless otherwise specified (see 6.2), the first article engine shall be subjected to a NATO 400-hour test. The test shall consist of four periods of 100 hours. Each 100-hour period shall consist of ten 10-hour cycles. Each 10-hour cycle shall be as specified in table V. The test shall be conducted under the following conditions:

a. Running time less than 30 minutes shall not be counted toward fulfillment of the 400 endurance hours.
b. The inlet air temperature shall be not less than 70°F nor more than 100°F.
c. The fuel used shall be as specified in 3.4.1.1.

TABLE V. Ten-hour cycle.

Period	% rated speed	% load	Duration (hours)
1	Idle	0	1/2
2	100	100	2
3	Governed speed	0	1/2
4	75	100	1
5	Idle _____ 100	0 _____ 100 4 min. 6 min.	2
6	60	100	1/2
7	Idle	0	1/2
8	Governed speed	70	1/2
9	Max. torque speed	100	2
10	60	50	1/2
		Total duration	10

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4.8.4.1.1.1 Power checks. Full rack performance characteristics of the engine shall be determined by operating the engine according to the following schedule before the NATO 400-hour test and after each 100-hour operating period:

Period	Engine speed (rpm)	Rack position
1	1600	Full rack.
2	2000	Full rack.
3	2400	Full rack.
4	2785	Full rack.

All data necessary to determine the performance characteristics specified in 4.8.4.1.1.3 shall be recorded in the log. Performance curves showing horse-power, torque, and specific fuel and oil consumption shall be required in addition to the log sheets. Oil consumption data shall be obtained from endurance hours.

4.8.4.1.1.1.1 Servicing. Prior to the power checks, the lubricating oil shall be changed, all filters serviced, and valve clearances checked and adjusted to applicable requirements.

4.8.4.1.1.1.2 Atmospheric conditions. The observed barometric pressure and wet and dry bulb temperatures existing within 4 feet of engine air inlet shall be recorded during the power checks.

4.8.4.1.1.2 Readings. The following readings shall be recorded at the end of each hour of operation, at the completion of each period, and just prior to stopping the engine (emergency stop excluded):

a. Engine speed, rpm.
b. Observed bhp.
c. Observed torque, pound-feet.
d. Air box pressure, inches Hg (gage).
e. Exhaust manifold pressure, inches of Hg (gage).
f. Fuel pressure after secondary filter, psi.
g. Fuel supply pressure at transfer pump, psi.
h. Lubrication oil pressure, gallery, psi.
i. Crankcase pressure, inches of water.
j. Lubricating oil temperature, sump, °F.
k. Lubricating oil temperature, inlet gallery, °F.
l. Fuel temperature after secondary filter, °F.
m. Air temperature at air cleaner inlet, °F.
n. Air temperature at air box, °F.
o. Coolant temperature, inlet and outlet, °F.
p. Exhaust temperature, °F.
q. Test cell ambient air temperature, °F.

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r. Blowby, cfm.
s. Fuel flow, pounds per hour.
t. Specific fuel consumption, pounds per bhp-hour.
u. Specific oil consumption, pounds per bhp-hour (at power check).
v. Barometric pressure 1/ of test cell - once each 4-hour period, inches Hg.
w. Exhaust smoke density (see 6.6.5).

1/ NOTE: Local observed barometric pressure.

4.8.4.1.1.3 Acceptable performance. During the NATO 400-hour test, the engine shall be observed for conformance to the following performance requirements during the power checks (see 4.8.4.1.1.1):

a. Speed range (see 3.4.2).
b. Governor (see 3.4.2.1).
c. Bhp (see 3.4.3).
d. Torque (see 3.4.4).
e. Exhaust smoke density (see 3.4.5).
f. Fuel consumption (see 3.4.6).
g. Oil consumption (see 3.4.7).
h. Oil pressure (see 3.4.8).
i. Engine life (see 3.4.1).

4.8.4.1.1.4 Teardown inspection. At the conclusion of the NATO 400-hour test, a teardown inspection as specified in 4.8.3.1 shall be performed.

4.8.4.2 Control test. Each engine selected in accordance with 4.6 shall be subjected to the 50-hour test specified in table VI. At each start, the engine shall be warmed up until the torque, speed, and temperatures have been stabilized for a period of not less than 1 minute. All applicable temperatures shall be recorded at the conclusion of each warm-up period. Warm-up starting and stopping times shall be recorded; however, warm-up time and running time of less than 30 minutes, unless specified in table VI, shall not be counted toward fulfillment of test run hours. Prior to the test, all applicable settings and adjustments (such as valve lash) shall be reset to specifications before the test is begun. The test shall be conducted under the following conditions:

a. The inlet air temperature shall be not less than 70°F nor more than 100°F.
b. The lubricating oil temperature shall not exceed 240°F.
c. The coolant temperature shall be maintained at 180 ± 5°F.
d. The fuel used shall conform to grade 2 of VV-F-800.

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TABLE VI. 50-hour test.

Period	Time (hrs)	Speed (rpm)	Rack position (% of full rack)
1	(run in schedule)	(see table I)	
2	5	1600	50
3	5	2000	75
4	5	2785	75
5	5	2400	100
6	30	2785	100
7	Power checks (see 4.8.4)		

4.8.4.2.1 Readings. The following readings shall be recorded after each hour of operation, at the completion of each period, and just prior to stopping the engine. All data shall be obtained under stabilized conditions, with the engine coupled to a dynamometer:

a. Engine speed, rpm.
b. Observed bhp.
c. Observed torque, pound-feet.
d. Air box pressure, inches Hg (gage).
e. Fuel pressure after secondary filter, psi.
f. Fuel supply pressure at inlet to engine driven supply pump, psi.
g. Lubricating oil pressure, psi.
h. Lubricating oil temperature, sump, °F.
i. Fuel temperature after secondary filter, °F.
j. Coolant temperature, inlet and outlet, °F.
k. Air temperature at air cleaner inlet, °F.
l. Cell ambient air temperature, °F.
m. Exhaust temperature, °F.
n. Fuel flow, pounds per hour.
o. Brake specific fuel consumption, pounds per observed bhp-hour.
p. Specific oil consumption, pounds per observed bhp-hour.
q. Exhaust smoke density (see 6.6.5).

4.8.4.2.2 Acceptable performance. The engine shall be observed for conformance to the following performance requirements at the completion of the 50-hour test, and shall be reinspected for the characteristics specified in table IV:

a. Speed range (see 3.4.2).
b. Governor (see 3.4.2.1).
c. Bhp (see 3.4.3).
d. Torque (see 3.4.4).
e. Exhaust smoke density (see 3.4.5).
f. Fuel consumption (see 3.4.6).
g. Oil consumption (see 3.4.7).
h. Oil pressure (see 3.4.8).

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4.8.5 Grades and slopes. To determine conformance to 3.4.9, the engine shall be operated for 30 minutes in each direction as specified in 3.4.9.

4.8.6 Steam and water jet cleaning. To determine conformance to 3.4.10, the engine shall be subjected to normal steam cleaning with cleaner conforming to P-C-437 at the completion of the first article endurance test and just prior to the teardown inspection. Immediately upon completion of the steam cleaning operation, the engine shall be subjected to normal water jet cleaning as specified herein. Prior to cleaning, the air intake opening, flywheel, exhaust, and all other openings that might be affected shall be covered. The engine shall start and operate at idle speed without misfiring within 2 minutes immediately after the completion of both the steam and water jet operations. During the disassembly of the engine, special notice shall be made of water entering the engine through joints, seals, gaskets, etc. Evidence of malfunction or of water seepage past joints, seals, gaskets, etc., shall constitute failure of the test.

4.8.7 Electromagnetic compatibility. Unless otherwise specified (see 6.2), conformance to 3.4.11 radiated emission suppression shall be determined by testing in accordance with MIL-STD-462, section 4.1 and 4.2, subsections 4.2.4 through 4.2.6.

4.8.8 Environmental.

4.8.8.1 Temperature.

a. Low. To determine conformance to 3.5.1, the engine, with integral winterization aids, shall be cold-soaked at an ambient temperature of -25°F for a period of 24 hours using the fuel specified in 3.4.1.1. The engine shall be started and shall operate for a sufficient time to stabilize engine operation.
b. High. To determine conformance to 3.5.1, the engine, without aids, shall be placed in a high-temperature chamber maintained at a temperature of 115°F and operated at full load until all temperatures are stabilized. The engine shall then be shut down, hot-soaked, and restarted when the fuel temperature at the inlet to the injection pump reaches its maximum value. Fuel shall be supplied to the engine mounted fuel supply pump at a pressure of 5 psi. The fuel pressure shall be shut off during the hot-soak period.

4.8.8.2 Elevation. To determine conformance to 3.5.2, the engine shall be placed in a simulated environmental condition of 8,000 feet elevation, a pressure of 22.2 inches Hg measured at the air cleaner inlet and exhaust outlet, and a temperature of 90°F. The engine shall be started and shall operate for a sufficient time to stabilize engine operation.

4.8.8.3 Humidity. To determine conformance to 3.5.3, the engine shall be placed in conditions as specified therein. The engine shall be started and shall operate for a sufficient time to stabilize engine operation.

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5. PACKAGING

5.1 Level of protection. Packaging shall be Level A or Level C as specified (see 6.2).

5.1.1 Level A. Engines entering the Military Supply Distribution System will be provided with Level A protection and shipped in a Long Life Reusable Container, PN11650202 (Cage Code 19207). Preservation, packaging, packing, and marking shall be in accordance with Special Packaging Instruction No. AK12957458 (PN57K1000, code ident 19207).

5.1.2 Level C. Engines being shipped from manufacturing site to production site will be provided with Level C protection. Preservation, packaging, packing, and marking shall be in accordance with MIL-STD-1190.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The engine covered by this specification is intended for production use, as a spare, or as a replacement for military tactical and combat vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, revision, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. If first article is required (see 3.1).
d. If responsibility for inspection and the place of inspection is other than the contractor's (see 4.1).
e. Type of first article sample, and the number of samples to be inspected (see 4.4 and 6.3).
f. First article inspection, if other than as specified (see 4.4).
g. Arrangements for first article inspection, approval of test results and disposition of first article (see 6.3).
h. Government's rights and conditions for waiving first article inspection (see 6.3).
i. Sample size for QCI examination and tests (see 4.5).
j. QCI acceptance criteria (see 4.5).
k. Control test sample size, frequency of tests, and disposition of lots covered by the sample size and of defective items (see 4.6).
l. Selection of applicable level and packaging requirements (see 5.1).
m. If all accessories and equipment shall not be installed on engines and adjusted (see 3.3.1).
n. If a log sheet is required, requirements for the log sheet (see 6.7).
o. Additional data for plates, if required (see 3.7).
p. If inspection conditions shall be other than as specified (see 4.3).
q. If testing for radiated emissions shall be other than as specified (see 4.8.7).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers on whether the engines should be preproduction samples, first article samples, or initial production samples, and for each inspection category, specify the number of samples to be inspected and the specific tests to be performed on each sample. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 Guidance for the contracting officer on applying first article. When a first article is required, consideration should be given to subjecting one engine produced under the production contract to first article inspection as specified in 4.4.

6.4 Guidance for the contracting officer on applying QCI sampling/ acceptance criteria.

6.4.1 QCI.

6.4.1.1 Sampling.

6.4.1.1.1 Lot formation. A lot should consist of all the engines from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

6.4.1.1.2 Sampling for examination. Consideration should be given to selecting samples for examination in accordance with general inspection level II of MIL-STD-105.

6.4.1.2 Examination.

6.4.1.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 6.4.1.1.2 should be examined for the defects listed in table IV to determine conformance to the following AQL's on the basis of percent defective.

Classification	AQL
Major	6.5
Minor	25

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6.4.1.3 Tests. Consideration should be given to subjecting each engine to the QCI tests specified in table III upon completion of the break-in run (see 3.3.3).

6.5 Guidance to the contracting officer on applying provisions for control tests. Consideration should be given to conducting control tests at the rate of 1 engine per month for production below 100 engines per month and 2 engines per month for production over 100 engines per month. Samples for control tests should have completed the examination of 4.5.1 and the tests of 4.5.2. The sample engine should be subjected to the control tests specified in table III beginning with the endurance test. After control testing, the engine should be examined for the defects specified in table IV.

6.6 Definitions.

6.6.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.6.2).

6.6.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.6.3 Bhp. Observed bhp shall be corrected to standard atmospheric conditions using the following formula:

$$\text{Corrected bhp} = \text{observed bhp} \times \frac{29.00}{B - e} \times \left(\frac{460 + t}{545}\right)^{1/2}$$

Where:

B = test barometric pressure, inches Hg wet.
e = Water vapor pressure, inches Hg.
t = intake air temperature, °F (see 5.2.2 of MIL-STD-1400).

6.6.4 Referee grade oil. The following designation applies for the grade required:

Grade	Government designation
15/40	MC 2777

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6.6.5 Exhaust smoke density. Exhaust smoke density should be determined with a Robert Bosch Model EFAW 68 Smoke Meter or equivalent (as approved by the Government) and Model EFAW 65 Sampling Pump or equivalent (as approved by the Government). The following may be used to visually define the degree of exhaust smoke density:

Description of exhaust smoke	Classification
Clear	1
Haze	2
Light gray	3
Medium gray	4
Dark gray to black	5

NOTE: Observation of exhaust should be made against a white background.

6.7 Log sheet. The contracting officer should specify in acquisition documents the Government's requirements for the log sheet, e.g., format, Government approval, and number of copies required.

6.8 Subject term (key word) listing.

Brake horsepower (bhp)
Compression-ignition
Endurance run
50-hour test
Governor
Internal-combustion
NATO 400-hour test
Torque

6.9 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Material Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

Custodian:
Army - AT

User activity:
DLA - CS

Preparing activity:
Army - AT

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